

ADVANCED MACHINE LEARNING · WORKSHEET 14

Maximum Likelihood Estimation and Logistic Regression

Probability · Likelihood · Sigmoid · Bernoulli Model · Binary Cross-Entropy

TEACHER COPY — Lecture 14

PART I: FROM PROBABILITY TO MAXIMUM LIKELIHOOD

SECTION 1 Begin With Probability: Outcome Uncertain, Parameter Fixed

A coin has probability $p = 0.7$ of landing heads. Let X be the number of heads in 10 independent tosses. The event is “exactly 7 heads.” In this probability question, the coin parameter is known and fixed, while the possible outcome is uncertain.

One particular sequence with 7 heads and 3 tails has probability $(0.7)^7(0.3)^3$. The binomial coefficient counts the positions in which the 7 heads can occur:

$$P(X = 7 \mid p = 0.7) = \binom{10}{7} (0.7)^7 (0.3)^3 \approx 0.2668.$$

Symbol	Meaning in this experiment
X	Number of heads observed in 10 tosses
p	Probability of heads on one toss; fixed at 0.7
$\binom{10}{7}$	Number of different positions in which 7 heads can occur
$P(X = 7 \mid p = 0.7)$	Probability of the outcome $X = 7$ when the coin parameter is known

◆ TAKEAWAY

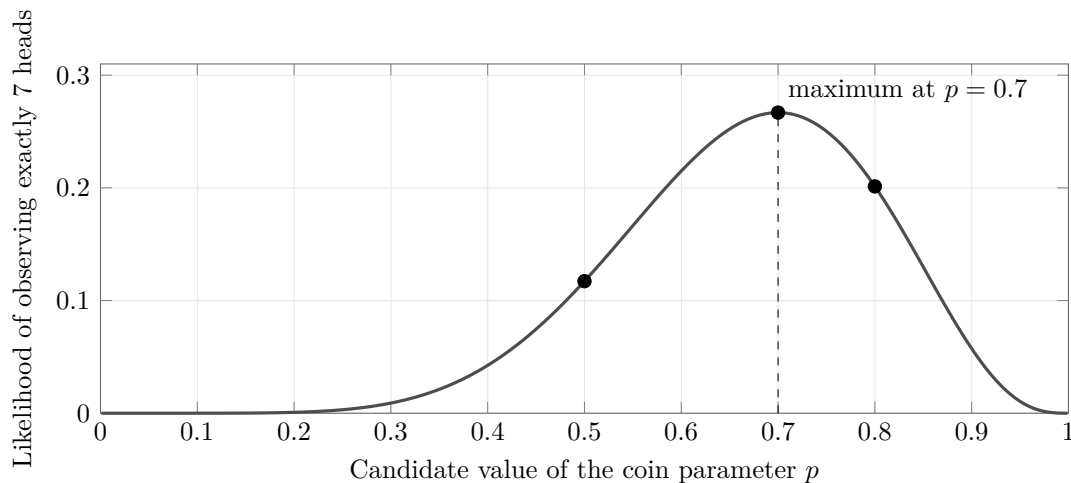
In a probability question, the model or parameter is fixed and the possible data or outcome varies.

SECTION 2 Turn the Question Around: Data Fixed, Parameter Varies

Now the observation is fixed: 7 heads were seen in 10 tosses. Candidate values of p are allowed to vary. The likelihood uses the same mathematical expression but asks which candidate coin parameter best explains the observed data:

$$L(p; X = 7) = \binom{10}{7} p^7 (1 - p)^3.$$

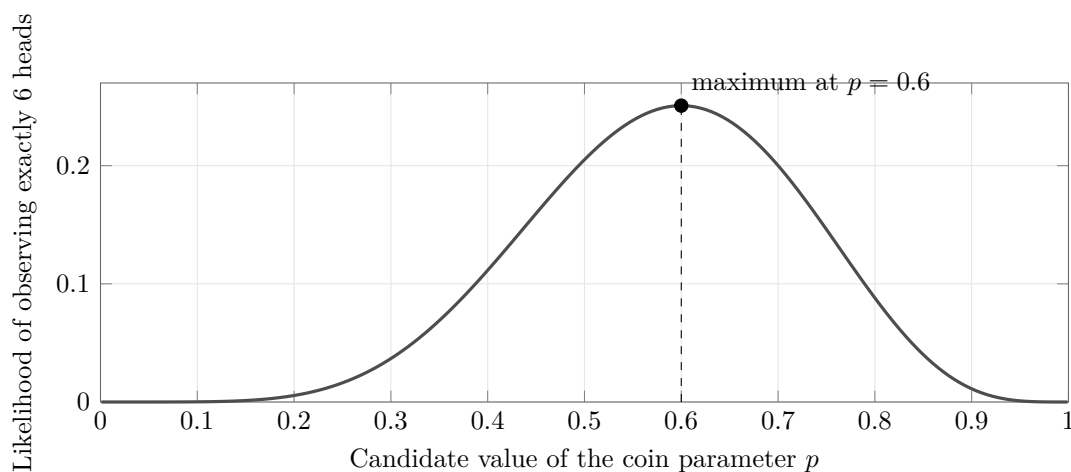
Candidate coin	Likelihood of 7 heads	Interpretation
$p = 0.5$	0.117	Possible, but provides weaker support for the observation.
$p = 0.8$	0.201	Explains the observation better than $p = 0.5$.
$p = 0.7$	0.267	Gives the highest likelihood across candidate p values.

Visual: The data stay fixed; the parameter changes

The likelihood function depends on the observations. If the experiment had instead produced 6 heads in 10 tosses, the fixed data would be $X = 6$ and the likelihood would become

$$L(p; X = 6) = \binom{10}{6} p^6 (1 - p)^4.$$

This is a different function of the candidate parameter p . Its maximum occurs at $p = 0.6$, so the peak shifts to the left of $p = 0.7$. Changing the observed number of heads changes which candidate value of p best explains the data.

Visual: With 6 heads observed, the likelihood peak moves left

SECTION 3 From Likelihood to Maximum Likelihood Estimation

Probability and likelihood have the same formula but different roles:

Question	Probability	Likelihood
What is fixed?	Model or parameter	Observed data
What varies?	Possible data or outcome	Candidate parameter values
Typical notation	$P(\text{Data} \mid \text{Parameter})$	$L(\text{Parameter}; \text{Data})$
Plain-language question	If this model is true, how likely is this outcome?	Given this outcome, which parameter explains it best?

Likelihood is a plausibility score over parameters; it is not automatically a probability distribution over them. For observed dataset D and parameter set θ , Maximum Likelihood Estimation chooses

$$\hat{\theta} = \arg \max_{\theta} L(\theta; D) = \arg \max_{\theta} P(D \mid \theta).$$

◆ TAKEAWAY

MLE chooses the parameters that make the data we actually observed most likely. It does not choose a “most probable parameter.”

PRACTICE P1 · PROBABILITY, LIKELIHOOD, AND MLE REASONING

- (a) The observed result is 7 heads in 10 tosses. Use the likelihood values above to compute

$$\frac{L(0.7; X = 7)}{L(0.5; X = 7)} = \text{approximately } 2.28, \quad \frac{L(0.7; X = 7)}{L(0.8; X = 7)} = \text{approximately } 1.33.$$

Interpret both ratios without treating likelihood as a probability distribution over p .

Expected answer: The observed 7 heads support $p = 0.7$ about 2.28 times as strongly as $p = 0.5$, and about 1.33 times as strongly as $p = 0.8$, among the candidate values being compared. The ratios compare plausibility scores; they are not posterior probabilities of the parameters.

- (b) Consider two investigations:

- A coin with known $p = 0.7$ has not yet been tossed; compare possible values of X .
- The result $X = 7$ has been observed; compare candidate values of p .

For each investigation, identify what is fixed and what varies, and explain why the second investigation is an MLE problem.

Expected answer: (i) p is fixed and the possible outcome X varies, so this is probability. (ii) $X = 7$ is fixed and candidate p values vary, so this is likelihood. MLE selects the candidate p that maximises that likelihood.

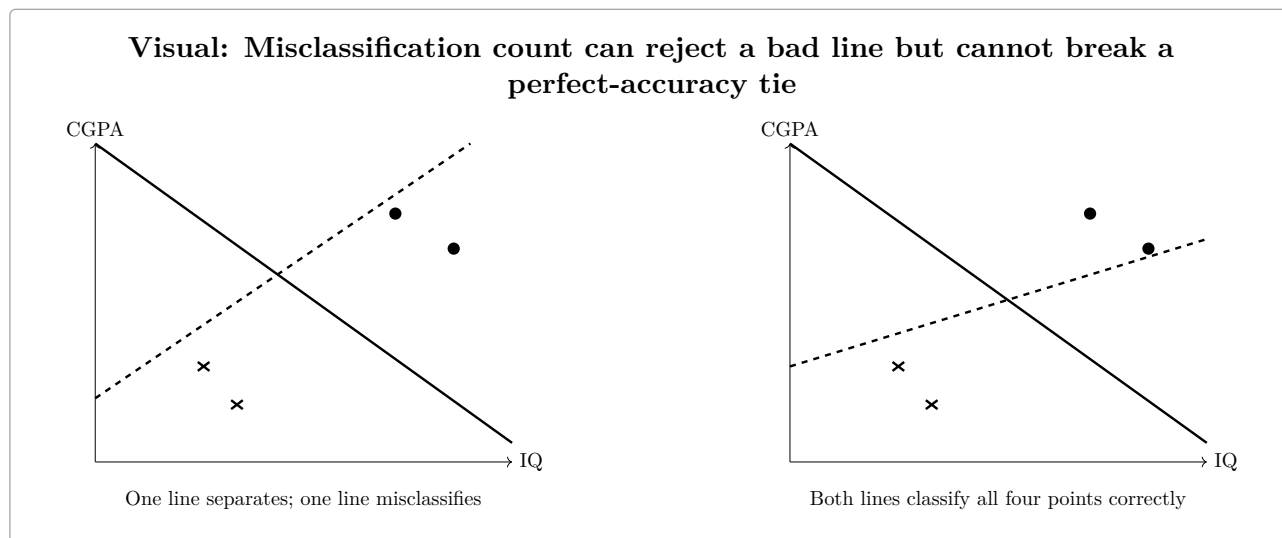
- (c) Candidate parameters θ_A and θ_B give likelihoods 0.18 and 0.24 for the same fixed dataset. Which candidate does MLE prefer? Would it be correct to claim that $P(\theta_B \mid D) = 0.24$? Justify both answers.

Expected answer: MLE prefers θ_B because $0.24 > 0.18$. No: the likelihood $P(D \mid \theta_B)$ is not automatically the posterior probability $P(\theta_B \mid D)$.

PART II: FROM A LINEAR SCORE TO A CLASS PROBABILITY

SECTION 4 Logistic Regression: Why Accuracy Alone Is Not Enough

Each student is described by IQ on the horizontal axis and CGPA on the vertical axis. The binary target is placement completed (1) or not completed (0). A straight line separates the feature space into two predicted classes.



Accuracy can reject a model that misclassifies points. But if two lines classify all four points correctly, both have zero misclassifications. A hard 0-or-1 score cannot prefer one parameter set, even when one boundary supports the observed labels more confidently.

★ KEY INSIGHT

The model needs a soft score that expresses how strongly it supports class 1 or class 0, is comparable across observations, and can be used inside a likelihood.

SECTION 5 The Linear Score and the Decision Boundary

For two features,

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

x_1 and x_2 are the features, β_1 and β_2 control their contributions, β_0 is the intercept, and z is a real-valued score rather than a probability.

The decision boundary is the set of points where $z = 0$. Points on one side have $z > 0$ and points on the other have $z < 0$. Changing β_0 shifts the line; changing feature weights changes its orientation and sensitivity.

Using $\beta_0 = -5$, $\beta_1 = 0.4$, $\beta_2 = 0.6$, and $(x_1, x_2) = (8, 7)$,

$$z = -5 + 0.4(8) + 0.6(7) = 2.4.$$

△ WARNING – SIGNED SCORE, NOT EXACT DISTANCE

Multiplying every weight, including the intercept, by the same positive constant leaves the line $z = 0$ unchanged but rescales z . The exact perpendicular distance is the score divided by the length of the feature-weight vector.

SECTION 6 Why a Hard Step Function Does Not Solve the Training Problem

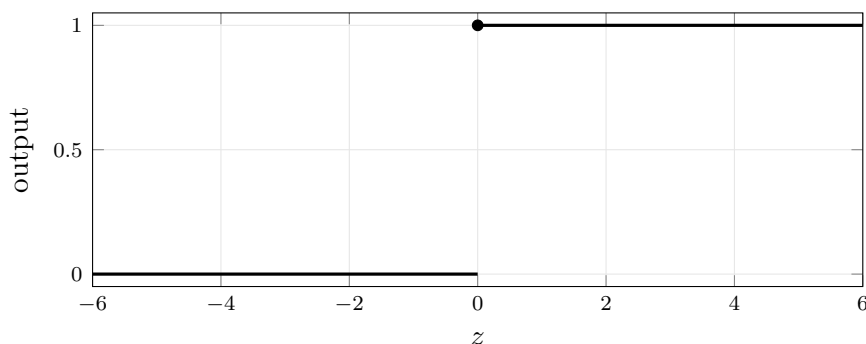
The hard decision rule is

$$\hat{y}_i = \begin{cases} 1, & z_i \geq 0, \\ 0, & z_i < 0. \end{cases}$$

It is useful when a categorical decision is eventually needed, but it is unsuitable as the probability model for fitting logistic regression:

- $z = 0.01$ and $z = 10$ both become class 1, so confidence is lost.
- Many boundaries can produce the same hard labels, so ties remain.
- The output is flat almost everywhere and jumps at zero, so small parameter improvements are not represented smoothly.

Visual: The step function keeps only a hard output

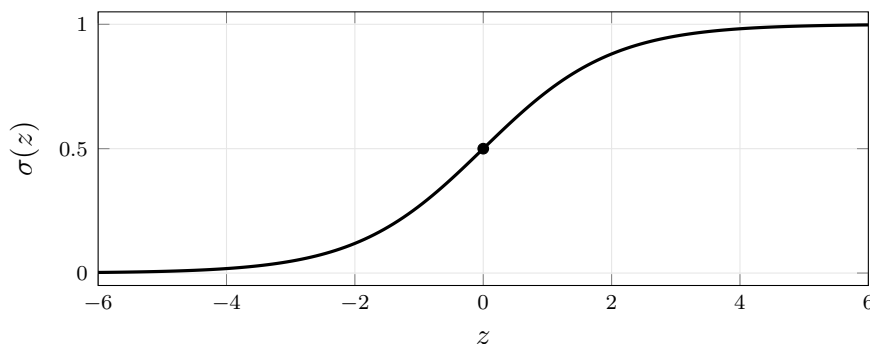


SECTION 7 The Sigmoid Function: Turn Any Score Into a Number Between 0 and 1

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

Because e^{-z} is positive, $0 < \sigma(z) < 1$. The sigmoid preserves the ordering of scores, changes smoothly, and has the range needed for a probability estimate.

Visual: The sigmoid turns the score into a smooth probability estimate



Score z	$\sigma(z)$	Class-1 interpretation
-4	0.018	Very weak support for class 1
-2	0.119	Low support for class 1
0	0.500	Exactly on the decision boundary
2	0.881	Strong support for class 1
4	0.982	Very strong support for class 1

As $z \rightarrow -\infty$, $\sigma(z) \rightarrow 0$; at $z = 0$, $\sigma(z) = 0.5$; and as $z \rightarrow +\infty$, $\sigma(z) \rightarrow 1$. The sigmoid output represents the model's estimated probability of class 1, not a guarantee that class 1 will occur.

SECTION 8 From Linear Score to Class Probability

For observation i ,

$$\hat{p}_i = P(Y_i = 1 \mid x_i; \beta) = \sigma(z_i).$$

The entire equation says that the model's estimated probability that observation i belongs to class 1 can be written in three equivalent ways: as $\hat{p}_i$, as the conditional probability $P(Y_i = 1 \mid x_i; \beta)$, or as the sigmoid of the score $\sigma(z_i)$.

Term	Meaning
$\hat{p}_i$	The probability of class 1 estimated by the model for observation i . The hat indicates that this value is produced by the fitted model.
$P(Y_i = 1 \mid x_i; \beta)$	The conditional probability that the response Y_i equals 1, given the feature vector x_i and the parameter vector β .
x_i	The complete feature vector describing observation i .
β	The model parameter vector containing the intercept and feature weights.
z_i	The linear score computed from x_i and β : $z_i = \beta_0 + \sum_{j=1}^d \beta_j x_{ij}$.
$\sigma(z_i)$	The sigmoid applied to the score. It converts any real-valued z_i into a number strictly between 0 and 1.

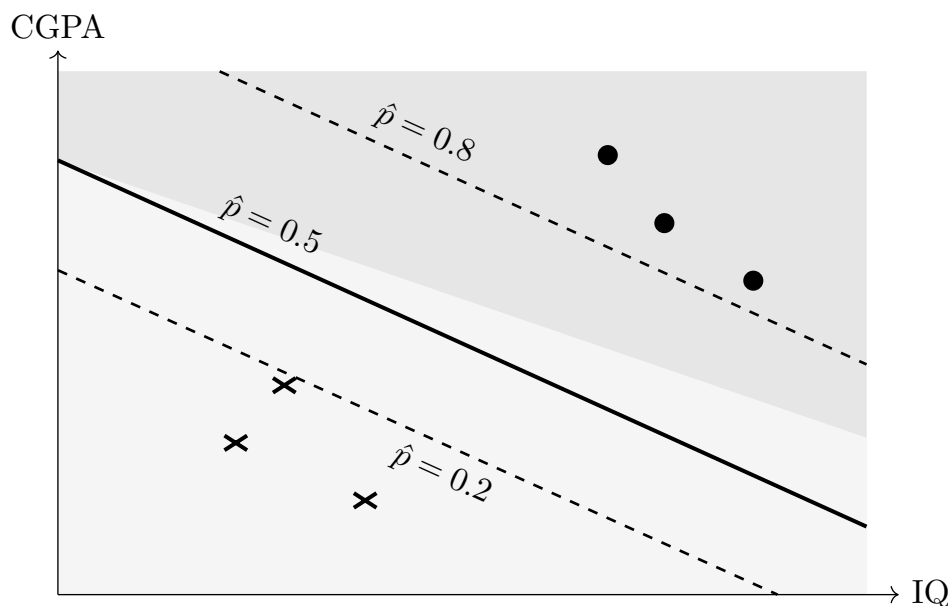
Because the two classes are complementary, the model-estimated probability of class 0 is

$$P(Y_i = 0 \mid x_i; \beta) = 1 - \hat{p}_i.$$

The two estimated probabilities therefore sum to 1: the probability assigned to class 0 is whatever remains after the model assigns probability $\hat{p}_i$ to class 1.

A boundary point has $z_i = 0$ and $\hat{p}_i = 0.5$. The probability changes nonlinearly as the score changes.

Visual: The line $\hat{p} = 0.5$ is one contour in a probability field



PRACTICE P2 · SCORE, BOUNDARY, SIGMOID, AND CONFIDENCE

(a) Using $\beta_0 = -5$, $\beta_1 = 0.4$, $\beta_2 = 0.6$, and $(x_1, x_2) = (8, 7)$:

- (i) Compute z : **2.4**.
- (ii) Compute $\hat{p} = \sigma(z)$ to three decimals: **0.917**.
- (iii) State the hard class at threshold 0.5: **class 1**.

(b) Multiply all three coefficients by 2.

- (i) Does the boundary $z = 0$ change? [**No**]
- (ii) Compute the new score and probability for the same point: $z' = \mathbf{4.8}$, $\sigma(z') = \mathbf{\text{approximately } 0.992}$.
- (iii) Explain why identical boundaries can still express different confidence.

Expected answer: Positive scaling leaves the zero set unchanged, so the boundary is identical. It increases the magnitude of nonzero scores, pushing sigmoid probabilities farther from 0.5 and therefore changing confidence and likelihood.

(c) Compare $z = 0.01$ and $z = 10$.

- (i) What hard label does the step function assign to each? **class 1 for both**.
- (ii) Their sigmoid values are approximately 0.5025 and 0.99995. Explain what the sigmoid preserves that the step function discards.

Expected answer: The sigmoid preserves strength of support: the first score is barely on the class-1 side, while the second indicates extremely strong class-1 support. The hard step collapses both to the same label.

PART III: FROM BERNOULLI LIKELIHOOD TO BINARY CROSS-ENTROPY

SECTION 9 Use Likelihood to Compare Candidate Logistic Models

For a positive observation, keep $\hat{p}_i$; for a negative observation, keep $1 - \hat{p}_i$. One compact actual-class probability is

$$q_i = \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}.$$

This is the standard Bernoulli probability formula written using the probability supplied by logistic regression. Its terms have the following roles:

- q_i is the probability that the model assigns to the label actually observed for data point i . With the data fixed, this is observation i 's contribution to the likelihood.
- $\hat{p}_i$ is the model-estimated probability of class 1 for observation i .
- $y_i \in \{0, 1\}$ is the actual observed class label.
- The exponents y_i and $1 - y_i$ act as selectors. Since any nonzero number raised to power 0 equals 1, one probability factor disappears and the other remains.

If $y_i = 1$, then $q_i = \hat{p}_i^1 (1 - \hat{p}_i)^0 = \hat{p}_i$. If $y_i = 0$, then $q_i = \hat{p}_i^0 (1 - \hat{p}_i)^1 = 1 - \hat{p}_i$. A Bernoulli observation can have only these two outcomes, with probabilities $\hat{p}_i$ and $1 - \hat{p}_i$. Therefore q_i is exactly the conditional Bernoulli probability of the outcome that occurred, and consequently the likelihood contribution of that observation when β is treated as the varying candidate parameter.

Observation	Actual y	Model A: probability of actual class	Model B: probability of actual class
Point 1	1	0.90	0.60
Point 2	1	0.80	0.55
Point 3	0	0.85 (= 1 - 0.15)	0.65 (= 1 - 0.35)
Point 4	0	0.75 (= 1 - 0.25)	0.60 (= 1 - 0.40)
Product	–	0.459	0.129

Both models have 100% training accuracy, but Model A assigns more probability to what actually happened, so its likelihood is larger.

SECTION 10 Formalise the Model: Notation and the Bernoulli Assumption

For n observations and d features,

$$z_i = \beta_0 + \sum_{j=1}^d \beta_j x_{ij} = \tilde{x}_i^\top \tilde{\beta}, \quad \hat{p}_i = \sigma(z_i), \quad Y_i | x_i \sim \text{Bernoulli}(\hat{p}_i).$$

Symbol	Meaning
n	Number of observations in the training dataset
d	Number of input features
x_i	Feature vector for observation i
y_i	Observed binary label for observation i
β	Parameter vector containing intercept and feature weights
z_i	Linear score for observation i
$\hat{p}_i$	Estimated probability that observation i belongs to class 1

The model assumes binary targets, conditional Bernoulli labels, conditionally independent observations, and log-odds linear in the features. “Success” is only Bernoulli terminology; it does not make class 1 practically or morally better.

△ WARNING – CONNECTION TO L13

Time-series observations may depend on earlier observations. The independent-product likelihood used here should not be copied blindly into a time-series setting.

SECTION 11 One Compact Formula Gives the Probability of the Observed Label

$$P(y_i | x_i; \beta) = \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}, \quad y_i \in \{0, 1\}.$$

Observed label	Substitution	Surviving probability
$y_i = 1$	$\hat{p}_i^1 (1 - \hat{p}_i)^0$	$\hat{p}_i$
$y_i = 0$	$\hat{p}_i^0 (1 - \hat{p}_i)^1$	$1 - \hat{p}_i$

If $\hat{p}_i = 0.8$, an observed label 1 contributes 0.8 to the likelihood, while an observed label 0 contributes 0.2. The formula selects the probability of the observed label, conditional on the feature vector.

SECTION 12 From One Observation to the Complete Dataset Likelihood

The complete dataset was observed as one joint event: observation 1 had its recorded label, observation 2 had its recorded label, and so on through observation n . Under the model’s conditional-independence assumption, the probability that all these label outcomes occur together is the product of their individual conditional probabilities. This is why the observation-level likelihood contributions are multiplied rather than added:

$$L(\beta) = \prod_{i=1}^n P(y_i | x_i; \beta) = \prod_{i=1}^n \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}.$$

Each factor measures how much probability the candidate model assigns to one observed label. The product $L(\beta)$ measures how strongly the same candidate parameter vector β supports the complete set of labels jointly. This multiplication is justified only after conditioning on the feature vectors and assuming the observations are independent under the model.

Changing β changes every score z_i , every probability $\hat{p}_i$, and therefore the product likelihood. MLE chooses

$$\hat{\beta} = \arg \max_{\beta} L(\beta).$$

Here β is a candidate vector of model parameters, $L(\beta)$ is the likelihood of the complete observed dataset produced by that candidate, and $\hat{\beta}$ is the selected maximum-likelihood estimate. The notation $\arg \max_{\beta}$ means “the value of β at which the following quantity is largest.” It returns the parameter vector that maximises the likelihood, not the numerical height of the likelihood itself. Thus the whole equation says: examine the possible parameter vectors and choose the one that makes the data actually observed most likely under the logistic-regression model.

The goal is not merely to place every $\hat{p}_i$ on the correct side of 0.5; it is to place as much probability as possible on the labels that were observed.

SECTION 13 Why We Take the Log of the Likelihood

A product of many probabilities can underflow. For example, 0.5^{2000} is around 10^{-602} , below the positive range represented by ordinary 64-bit floating-point arithmetic.

Because logarithm is strictly increasing, it preserves the parameter that maximises the objective. It also converts products into sums:

$$\log\left(\prod_{i=1}^n q_i\right) = \sum_{i=1}^n \log(q_i).$$

Therefore the logistic-regression log-likelihood is

$$\ell(\beta) = \log L(\beta) = \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)].$$

Log-likelihood values are usually non-positive; a value closer to zero is larger and therefore better.

SECTION 14 Convert Maximum Log-Likelihood Into Binary Cross-Entropy

Maximising log-likelihood is equivalent to minimising its negative. Dividing by n changes the scale from a total to an average without changing the best β for a fixed dataset:

$$\text{BCE}(\beta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)].$$

For one observation,

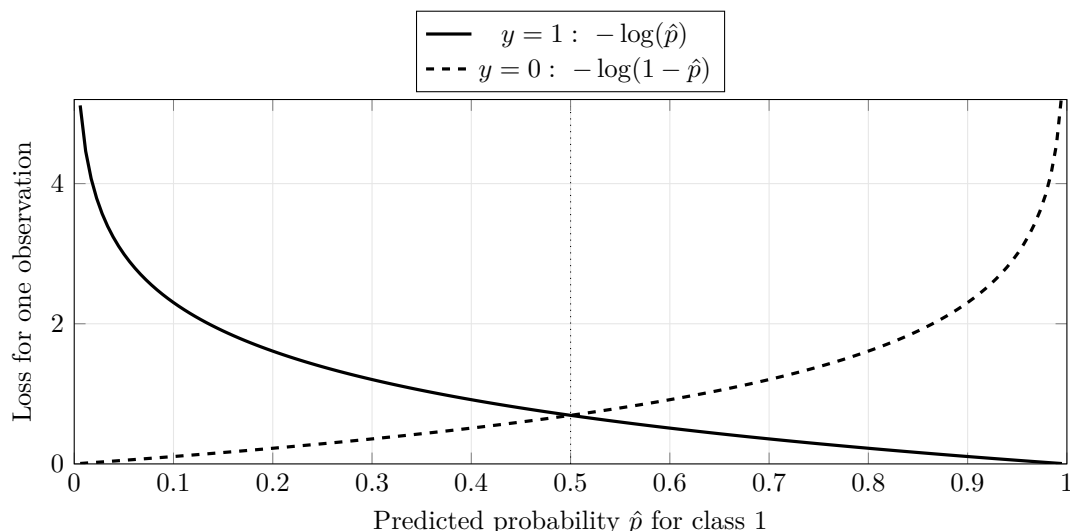
$$L_i = \begin{cases} -\log(\hat{p}_i), & y_i = 1, \\ -\log(1 - \hat{p}_i), & y_i = 0. \end{cases}$$

★ KEY INSIGHT

Binary Cross-Entropy is the negative average Bernoulli log-likelihood produced by the sigmoid probability model.

SECTION 15 How Binary Cross-Entropy Behaves

The horizontal axis in the graph is the model's predicted probability $\hat{p}$ for class 1. The vertical axis is the BCE loss for one observation. The solid curve applies when the actual label is $y = 1$, and the dashed curve applies when the actual label is $y = 0$. At any chosen value of $\hat{p}$, the appropriate curve gives the loss after the true label is known.

Visual: BCE rewards correct confidence and punishes wrong confidence

Correct confidence means assigning high probability to the class that actually occurred: $\hat{p}$ close to 1 when $y = 1$, or $\hat{p}$ close to 0 when $y = 0$. In either case, the probability of the observed class is close to 1, its negative logarithm is close to 0, and BCE is low.

Wrong confidence means assigning very little probability to the class that actually occurred: $\hat{p}$ close to 0 when $y = 1$, or $\hat{p}$ close to 1 when $y = 0$. As the probability assigned to the observed class approaches 0, the negative logarithm grows without bound. This is why a confidently wrong prediction receives a very large loss rather than merely the same penalty as a slightly wrong prediction.

For $y = 1$, predictions 0.90, 0.60, and 0.10 give losses 0.105, 0.511, and 2.303. The solid curve therefore falls as the class-1 probability increases. For $y = 0$, the pattern mirrors around 0.5: $\hat{p} = 0.10$ gives 0.105, while $\hat{p} = 0.90$ gives 2.303, so the dashed curve rises as the class-1 probability increases. The graph directly connects each predicted probability to the penalty appropriate for the corresponding actual label.

◆ TAKEAWAY

Accuracy only asks whether a probability crossed a threshold. BCE asks how much probability the model assigned to the truth.

PRACTICE P3 · LIKELIHOOD, LOG-LIKELIHOOD, AND BCE

- (a) Use the four-observation Model A and Model B table in Section 9.
 - (i) Recompute each product: $L_A = 0.459$ and $L_B = 0.1287 \approx 0.129$.
 - (ii) Compute log-likelihoods to three decimals: $\ell_A = -0.779$; $\ell_B = -2.050$.
 - (iii) Compute average BCE to three decimals: $\text{BCE}_A = 0.195$; $\text{BCE}_B = 0.513$.
 - (iv) Explain why MLE and BCE choose the same model.

Expected answer: Model A has the larger likelihood and log-likelihood. Negating and averaging reverses maximisation into minimisation, so Model A also has the smaller BCE.

- (b) A model outputs $\hat{p} = 0.9$. Compute the Bernoulli likelihood contribution and per-observation BCE for:
 - (i) $y = 1$: contribution = 0.9, loss = **approximately 0.105**;
 - (ii) $y = 0$: contribution = 0.1, loss = **approximately 2.303**.

Explain why the second case is penalised much more strongly.

Expected answer: The second case assigns only 0.1 probability to the label that actually occurred. The negative logarithm converts that very small support into a large penalty.

(c) Starting from

$$L(\beta) = \prod_{i=1}^n \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i},$$

show the two algebraic transformations that produce BCE. State why neither transformation changes the best β for a fixed dataset.

Expected answer: Take log to convert the product to $\sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$; strict monotonicity preserves the maximiser. Then negate and divide by n to obtain BCE; negation changes maximisation to minimisation and division by fixed positive n only rescales the objective.

(d) If one actual-class probability is exactly zero, what happens to the product likelihood, log-likelihood, and BCE? Connect the result to numerical clipping or stable implementations.

Expected answer: The product likelihood becomes zero; its logarithm tends to $-\infty$; the negative log-loss tends to $+\infty$. Since $\log(0)$ is undefined numerically, implementations use stable formulas or keep probabilities away from exactly 0 and 1.

PART IV: WHAT THE OBJECTIVE MEANS FOR TRAINING

SECTION 16 What the Derived Objective Tells Us About Training

For standard binary logistic regression, BCE is convex as a function of the weights. Informally, the objective has a bowl-like global structure with no separate bad local minima. Convex does not mean quadratic, and it does not imply that a direct formula exists.

Unlike the normal equation for linear regression, the sigmoid couples weights to probabilities nonlinearly, so the optimal logistic-regression weights cannot generally be isolated in one closed-form expression. An iterative optimization algorithm is required.

The BCE derived here is the data-fit term. L1 or L2 regularization can be added:

$$J(\beta) = \text{BCE}(\beta) + \lambda R(\beta),$$

where $R(\beta)$ is an L1 or L2 penalty and λ controls its strength.

△ WARNING – DO NOT DERIVE TODAY

Gradient calculations, parameter-update equations, and multiclass logistic regression belong to L15. Lecture 14 stops after establishing the probability model, the BCE objective, and why iterative optimization is needed.

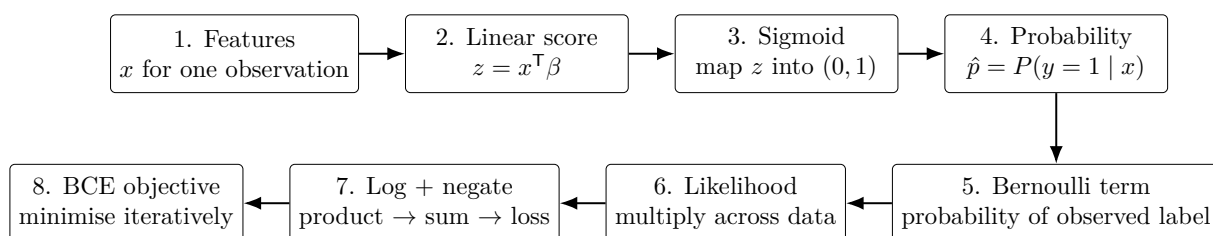
L14 establishes	L15 will add
$\hat{p} = \sigma(x^\top \beta)$	Derivative of BCE with respect to weights
Objective to minimise: BCE	Gradient Descent update steps and implementation
Binary labels 0 and 1	Extension to multiclass classification

SECTION 17 Misconception Checks and Assessment-Ready Prompts

Misconception	Correction
“Likelihood is $P(\text{parameter} \mid \text{data})$.”	Likelihood is $P(\text{data} \mid \text{parameter})$ viewed as a function of the parameter.
“The sigmoid output is the final class.”	It is an estimated probability; a threshold may later convert it to a class.
“Any correct prediction gets zero loss.”	A barely correct probability still receives positive BCE; confidence matters.
“BCE is just a formula to memorise.”	It is the negative average Bernoulli log-likelihood.
“Convex means a direct formula exists.”	Convexity describes objective shape; logistic regression still needs iterative optimization.

SECTION 18 End the Lecture as One Continuous Story

Visual: Complete logic of binary logistic regression from features to BCE



1. Training data are observed, so parameters are judged by how well they explain those data.
2. A linear combination of features gives a score z and boundary $z = 0$.
3. A step function loses confidence information; sigmoid maps z to $\hat{p}$.
4. A Bernoulli term selects the probability of the observed label.
5. Conditional independence produces the dataset likelihood product.
6. Taking log gives a stable sum without changing the best parameters.
7. Negating and averaging produces BCE, the objective to be minimised iteratively.

◆ TAKEAWAY

Probability $\rightarrow$ Likelihood $\rightarrow$ MLE $\rightarrow$ Linear score $\rightarrow$ Sigmoid $\rightarrow$ Bernoulli PMF $\rightarrow$ Dataset likelihood $\rightarrow$ Log-likelihood $\rightarrow$ BCE.

PRACTICE P4 · CONCEPTUAL AND MATHEMATICAL SYNTHESIS

- (a) Two models both classify a positive example correctly. Model A predicts 0.51 and Model B predicts 0.95.
- (i) Compute BCE for each: A = **approximately 0.673**, B = **approximately 0.051**.
 - (ii) Which model is preferred, and why?
Expected answer: Model B is preferred for $y = 1$ because it assigns much more probability to the observed positive label and therefore has lower BCE.
 - (iii) If the actual label is instead 0, compute the losses and reverse the comparison.
Expected answer: A gives $-\log(0.49) \approx 0.713$; B gives $-\log(0.05) \approx 2.996$. Model A is now preferred because it assigns more probability to class 0.

- (b) Compare log-likelihood values -35 and -20 . A student selects -35 because its magnitude is larger. Diagnose the reasoning.

Expected answer: Log-likelihood is maximised, so the larger numerical value is better. Since $-20 > -35$, -20 is better; magnitude is not the comparison rule.

- (c) A time-series dataset is inserted directly into the independent-product Bernoulli likelihood. What assumption must be examined before accepting the derivation, and why?

Expected answer: Examine conditional independence. Time-series labels may depend on earlier observations, so multiplying observation-level probabilities as if they were conditionally independent may be invalid.

- (d) Evaluate each claim with a mathematical or model-based justification.

- (i) “If two models have the same accuracy, they must have the same likelihood.” [F]

Expected answer: Equal hard labels do not imply equal probabilities for the observed labels; their likelihood products can differ.

- (ii) “Taking log can change which parameter maximises likelihood.” [F]

Expected answer: Log is strictly increasing, so it preserves the ordering and the maximiser.

- (iii) “Convex BCE guarantees a closed-form solution.” [F]

Expected answer: Convexity concerns objective shape, not algebraic solvability; the sigmoid coupling still requires iterative optimization.

- (iv) “Adding $\lambda R(\beta)$ changes the objective by penalising weights in addition to fitting the data.” [T]

Expected answer: BCE remains the data-fit term, while the L1 or L2 regularizer discourages extreme weights according to λ .

- (e) Reconstruct the full derivation by writing one equation at each arrow:

$$x_i \longrightarrow z_i \longrightarrow \hat{p}_i \longrightarrow P(y_i \mid x_i; \beta) \longrightarrow L(\beta) \longrightarrow \ell(\beta) \longrightarrow \text{BCE}(\beta).$$

Do not compute gradients or parameter updates.

Expected answer: $z_i = \beta_0 + \sum_j \beta_j x_{ij}$; $\hat{p}_i = \sigma(z_i)$; $P(y_i \mid x_i; \beta) = \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i}$; $L(\beta) = \prod_i P(y_i \mid x_i; \beta)$; $\ell(\beta) = \sum_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)]$; $\text{BCE}(\beta) = -\ell(\beta)/n$.

BY THE END OF THIS SHEET, YOU CAN

- ☐ Distinguish probability from likelihood by identifying what is fixed and what varies.
- ☐ Explain MLE as a rule for choosing parameters from observed data.
- ☐ Separate a linear score, the boundary $z = 0$, and a final hard class.
- ☐ Explain why the hard step function is unsuitable for likelihood-based fitting.
- ☐ Use the sigmoid to convert a score into an estimated probability.
- ☐ Write the Bernoulli probability of an observed binary label.
- ☐ Build the independent dataset likelihood and transform it into log-likelihood.
- ☐ Derive BCE as negative average Bernoulli log-likelihood.
- ☐ Explain how BCE reacts to correct, uncertain, and confidently wrong predictions.
- ☐ Explain why logistic regression requires iterative optimization without deriving the L15 updates.

ANSWER KEY AND MARKING GUIDE

P1. Likelihood ratios: 2.28 and 1.33. Probability fixes the parameter and varies the possible outcome; likelihood fixes the observation and varies candidate parameters. MLE prefers the larger likelihood, but likelihood is not a posterior probability over parameters.

P2. $z = 2.4$, $\hat{p} \approx 0.917$, class 1. Positive scaling leaves $z = 0$ unchanged but moves sigmoid probabilities away from 0.5. The step function gives class 1 for both $z = 0.01$ and $z = 10$, while the sigmoid distinguishes weak from strong support.

P3. $L_A = 0.459$, $L_B = 0.1287$; $\ell_A \approx -0.779$, $\ell_B \approx -2.050$; $\text{BCE}_A \approx 0.195$, $\text{BCE}_B \approx 0.513$. For $\hat{p} = 0.9$, the contributions for $y = 1$ and $y = 0$ are 0.9 and 0.1, with losses 0.105 and 2.303. Log converts product to sum; negate and average to get BCE. Zero actual-class probability gives zero likelihood and unbounded log-loss.

P4. At $y = 1$, losses for 0.51 and 0.95 are 0.673 and 0.051; at $y = 0$, they are 0.713 and 2.996. Log-likelihood -20 is better than -35 . The independent-product assumption must be checked for time series. Equal accuracy need not mean equal likelihood; log preserves the maximiser; convexity does not imply closed form; regularization adds a weight penalty. The final equation chain ends with $\text{BCE}(\beta) = -\ell(\beta)/n$.